Interview Prep Real Estate

- Educational Presentation #18 – Real Estate
- 13.02.2026
- Property of the Alternative Investment Society (AIS)
- Sources: WSP, REIA

ESCP Alternative Investment Society

Section I: Industry Fundamentals

Q: What are the 4 types of Real Estate Strategies?

- There are 4 key strategies within real estate investing, differentiated based on their risk/return profile

Low Risk/Return → **High Risk/Return**

Core

- Core funds target a **7-10%** return through acquiring prime properties which have long-term leases with high-quality tenants
- There is high conviction of stable income, and low uncertainty

Core+

- Core+ take on more risk, aiming for returns between **10-12%**
- Projects taken on by Core+ funds often require capital expenditure to optimize cash-flow, increasing risk

Value-add

- At this level, funds are looking for return around **12-15%**, as projects require major capex spending, after which rents hope to be drastically increased
- Often in sup-prime locations

Opportunistic

- Opportunistic investors require a **15-20%** return to compensate for major risk of undertaking complicated projects
- Often these projects have no initial positive cash-flow

Q: What is the Real Estate Capital Stack?

- The real estate capital stack is comprised of debt as well as equity of several types, differentiated by their risk-return profiles
- The common capital stack includes senior debt and mezzanine debt with clear priority rights; however, a variation is to structure senior debt into an A-note (senior) and B-note (subordinated), with different risk-return profiles

Senior Debt	Comprises the largest portion of the capital, gives lowest yield for lenders, is drawn last in capital deployment and has first repayment priority (underlying RE and CFs as collateral)
Mezzanine Debt	Yields for subordinate debt are typically higher given its secondary collateral and lower repayment priority, meaning more risk for private lenders
Preferred Equity	Preferred equity offers the investor a stake in the upside of the property, whilst also receiving preferential repayment conditions to reduce risk compared to sponsor equity
Common Equity	Common equity (sponsor) holders have the highest potential upside in a project; however, they also carry the highest risk with the lowest repayment priority in case of default

Low risk-return

High risk-return

Q: What are current Threats and Opportunities in the RE Market?

Threat

1	Higher Financing Costs
2	Persistent Macro Risk
3	Rising Construction & Operating Costs
4	Regulatory Constraint

Opportunity

1	Sustained Demand & Rental Growth
2	Institutional Capital Flows
3	Regional and Sector Diversification
4	ESG Investments

Interview Answer

”Opportunities in the current real estate market include sustained demand and growth for certain markets such as secondary cities, data centers and electricity generation especially if they integrate ESG into their core design. Threats mostly come from increasing costs whether in the financing stage with higher interest rates or in the construction and operation stage with increasing costs for labor and material.”

Q: What are some of the major types of RE firms?

Private Equity / Real Estate Funds

- They raise money from institutional investors to acquire and manage properties
- Value is created through repositioning, operational improvements, strategic exits, among other things

REITs (Real Estate Investment Trust)

- Often publicly traded companies (though they can also be private vehicles) that operate income-producing real estate
- In the U.S. REITs must distribute at least 90% of taxable income as dividends, to benefit from pass-through taxation

Developers

- Buys land and oversees construction or redevelopment projects
- Value created through leasing, stabilization of property NOI, and eventual exit

Debt Providers

- Do not get involved in owning or building the property, but instead provides the financing through mortgages, mezzanine loans, construction financing etc.
- Earn returns through interest payments and structured credit

Interview Answer

"In real estate, firms can broadly be grouped into equity investors such as private equity funds and REITs, developers who create and reposition assets, and debt providers who finance transactions. Additionally, the ecosystem features brokers, operators, and advisory firms which support difference areas in the value chain."

Q: What are the Different Real Estate Asset Classes?

- Properties are categorized into asset classes by function and operating characteristics, some of which are shown below
- Within these asset classes, properties are also assigned a grade (Class A-C) to show relative quality and expected risk-return profile

Office	• Primarily leased to corporate tenants on medium to long term leases (5-10+ years) • Cash flow is sensitive to vacancy and economic cycles
Retail	• Includes shopping centers and high-street retail with consumer facing tenants • Performance driven by tenant sales and location quality
Industrial	• Warehouses and logistics facilities supporting distribution and e-commerce • Typically, lower capex and strong demand from supply chains
Multifamily	• Residential rental properties with short-term leases • Stable demand base with consistent cash flow turnover
Hospitality	• Hotels generating revenue from nightly room rates (ADR x occupancy) • Most are operationally intensive and highly cyclical

Section II: Valuation and Deal Analysis

Q: What are the Primary Methods for Property Valuation?

- Properties are mostly valued using three approaches: sales comparison, income-based valuations, and cost-based valuations
- The appropriate method depends on the property's characteristics, income profile, and availability of market data

Sales Comparison Approach

- Values a property based on the recent transactions of comparable assets ("comps")
- Adjustments are made for differences in location, size, condition, and tenancy
- Most effective in markets with high transaction volume and reliable data for comparison

Income Approach

- Values property based on forecasted future cash flows
- Includes Direct Capitalization (NOI/Cap Rate), Discounted Cash Flow (DCF), and Gross Rent Multiplier (GRM)
- Most common method for income-producing and institutional investment properties

Cost Approach

- Value is determined by calculating land value plus replacement cost of improvements, less depreciation
- Often used for new developments, special use properties, or assets with limited comparable sales
- Less common for stabilized, income-producing investment assets

Q: How are Cap Rates used in Valuations?

- Cap rate, or capitalization rate, is one of the main financial metrics used for valuation of properties
- It is based on comparing the expected net operating income (NOI) to the purchase price

Cap Rate (%) = Net Operating Income (NOI) / Purchase Price

Cap Rates in Valuation

- On the sell side, by looking at cap rates of comparable properties (similar asset class and geography), investors can assess what the market value of their own property is, by combining the market cap rate with their asset NOI
- On the buy side, investors use cap rate as a measure of how risky the asset is to hold, as well as compare it to similar properties to make sure they are not overpaying
- Cap rates differ drastically between geographies and asset classes

Interpretation of Cap Rates

- A higher cap rate may indicate higher risk but potentially higher returns, while a lower cap rate implies the opposite
- To successfully value an investment property, cap rates must be combined with other investment considerations

Q: What are cap rates of different asset classes relative to each other?

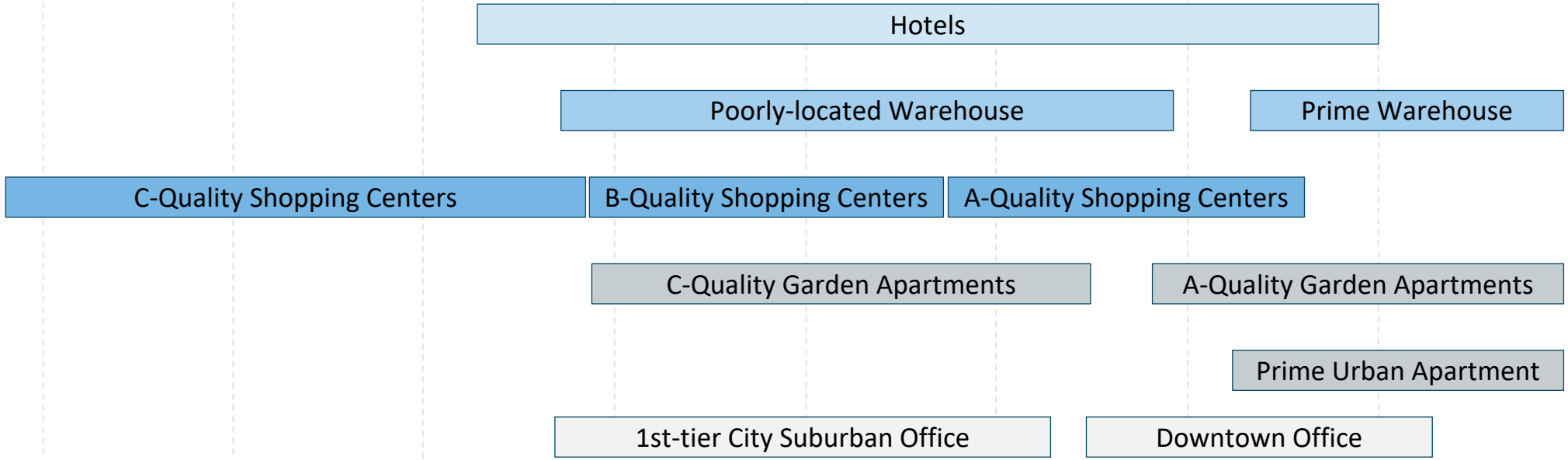
Cap Rate Spectrum

12%	11%	10%	9%	8%	7%	6%	5%	4%

Equivalent Income Multiples

8.3x	9.1x	10.0x	11.1x	12.5x	14.3x	16.7x	20x	25x

Relatively high Cap Rate: Less expensive per dollar of NOI *Relatively low Cap Rate: More expensive per dollar of NOI*



Note: The following cap rates are an estimation from research, however, large discrepancies exist in the real world and for each market

ESCP Alternative Investment Society

Q: Two properties have the same fundamentals, why does value differ?

Aspects that may affect value

1	Location Micro Differences
2	Tenant Quality
3	Lease Structure
4	Vacancy
5	Zoning Upside
6	Ownership & Transaction Context
7	Market Perception & Branding
8	Capital Structure

Interview Answer

"Two buildings right next to each other can have differences in value due to multiple factors. The most common is slight differences in locations such as a better view or access to transport entrances, for example being in center of a development versus a corner block. These differences exist even if the property was developed by the same developer. Moreover, if the two buildings are by different developers the branding or financing structure may differ, and this will change the rent they can charge. Hence, NOI will not be the same resulting in a higher or lower property value."

Value = NOI/Cap Rate

Q: Does Student Housing or Multi-family have a higher cap rate? Why?

- When analyzing the cap rate of one asset class relative to another we identify the characteristics of each asset class, to then determine what risk/return profiles they have
- We assume in this problem that the properties are of the same location and quality as the other, and that the only difference is in use case

Student Housing:

- University-dependent
- By-the-bed in 12-month cycles
- High operational intensity and annual turnover
- Seasonal stability

→ Higher perceived risk

Multi-family:

- Broad rent pool
- Rented by unit with rolling leases
- Moderate operational intensity and annual turnover
- Very stable income

→ Lower perceived risk

- The lower perceived risk of multi-family, all other things considered equal will lead it to have a lower cap rate than student housing

Interview Answer

"All other things considered equal, student housing properties have more inherent risk than multi-family. This is due to a smaller pool of potential tenants, higher annual turnover and higher operational intensity in student housing. An increased risk must be rewarded with higher returns, leading student housing generally to have a higher cap rate."

Section III: DCF and Financial Modelling

Q: Walk me through the steps to create a Real Estate DCF (1/2)

1

Gross Potential Income

Firstly, model what the income of the property will be assuming 100% occupancy, which can be based on current rent or the rent you plan to lease at, including indexation over time (increase in rent year-over-year, usually CPI)

2

Vacancy Loss

Properties rarely operate at 100% occupancy, meaning that present and future vacancy loss, as well as future downtime cost between leases must be accounted for to arrive at the effective gross income (EGI)

3

Operating Expenses, Asset Management fees and Taxes

To arrive at the NOI you deduct operating expenses (if no triple-net (NNN) lease is in place) including repairs and periodic maintenance estimated to grow with inflation, asset management fees and property taxes from the EGI

Interview Answer

"I'd start by projecting Gross Potential Income assuming full occupancy and applying rent growth. Then I'd deduct vacancy and credit loss to arrive at Effective Gross Income. From there, I'd subtract operating expenses, property taxes and management fees to calculate NOI."

Q: Walk me through the steps to create a Real Estate DCF (2/2)

4

Cash-Flow Attributable to Equity

From the NOI you subtract the cost of the debt (1) taken on the property (interest and principal payments), as well as potential tenant improvement costs (2) and capex (3), to arrive at the income attributable to equity

5

Sale Proceeds

The terminal value represents most of the total project value and is calculated by dividing final year NOI with an estimated exit cap rate, after which the sale cost and loan payoff is deducted to arrive at the sale proceeds

6

Metric Calculation

When the yearly cash-flows have been calculated, along with the final year sale proceeds, the IRR (unlevered and levered), Roe, EM along with other metrics can be calculated, determining the attractiveness of the investment

Interview Answer

"Next, I'd subtract debt service, capital expenditures, and any tenant improvement or leasing costs to arrive at cash-flow to equity. In the final year, I'd calculate the terminal value using an exit cap rate applied to stabilized NOI, deduct selling costs and remaining loan balance to determine sale proceeds. Finally, I'd calculate the return metrics."

Q: Differentiate between Cash-on-cash, IRR and NOI Yield?

- The Cash-on-cash, IRR and NOI Yield are all metrics used to evaluate the return on a real estate asset, but differ in time horizon, leverage impact as well as accounting for time value of money

Cash-on-Cash	IRR	NOI Yield
• Short term, annualized metric • Measures annual income of the property compared to equity invested • No account of the time value of money, but accounts for financing after debt service, as the base is invested equity **Cash-on-Cash = Annual Pre-Tax Cash Flow / Total Equity Invested**	• Multi-year performance metric • Counts the annualized return over the entire asset holding period • Accounts for the time value of money, hence sensitive to timing, and the sale proceeds $0 = \Sigma\ (\text{Cash Flow}_t / (1 + \text{IRR})^t)$	• Property level metric, before financing • Gives the unlevered property income relative to its value, or to its purchase price, depending on how it is used • Like the cap rate, and is used to compare assets **NOI Yield = NOI / Property Value (or Purchase Price)**

Q: What are the Metrics to Assess a Property's Ability to Service Debt?

- Lenders evaluate a property's ability to support financing by analyzing cash flow coverage, leverage levels, and income-based yield to decide whether cash flow is adequate to meet obligations and whether loan is appropriately sized relative to risk

Loan-to-Value (LTV)

- Shows how much debt is backed by the property's value, helping lenders assess downside protection

LTV (%) = Loan Amount / Appraised Property Value

Loan-to-Cost (LTC)

- Measures how much a project is financed with debt, showing risk exposure prior to stable income

LTC (%) = Loan Amount / Total Project Cost

Debt Service Coverage Ratio (DSCR)

- Used to determine borrower's ability to pay based on how many times the earnings cover the debt service

DSCR (x) = NOI / Annual Debt Service

Debt Yield

- Evaluates the return to the lender of the loan, based on the NOI relative to the loan amount

Debt Yield (%) = NOI / Total Loan Amount

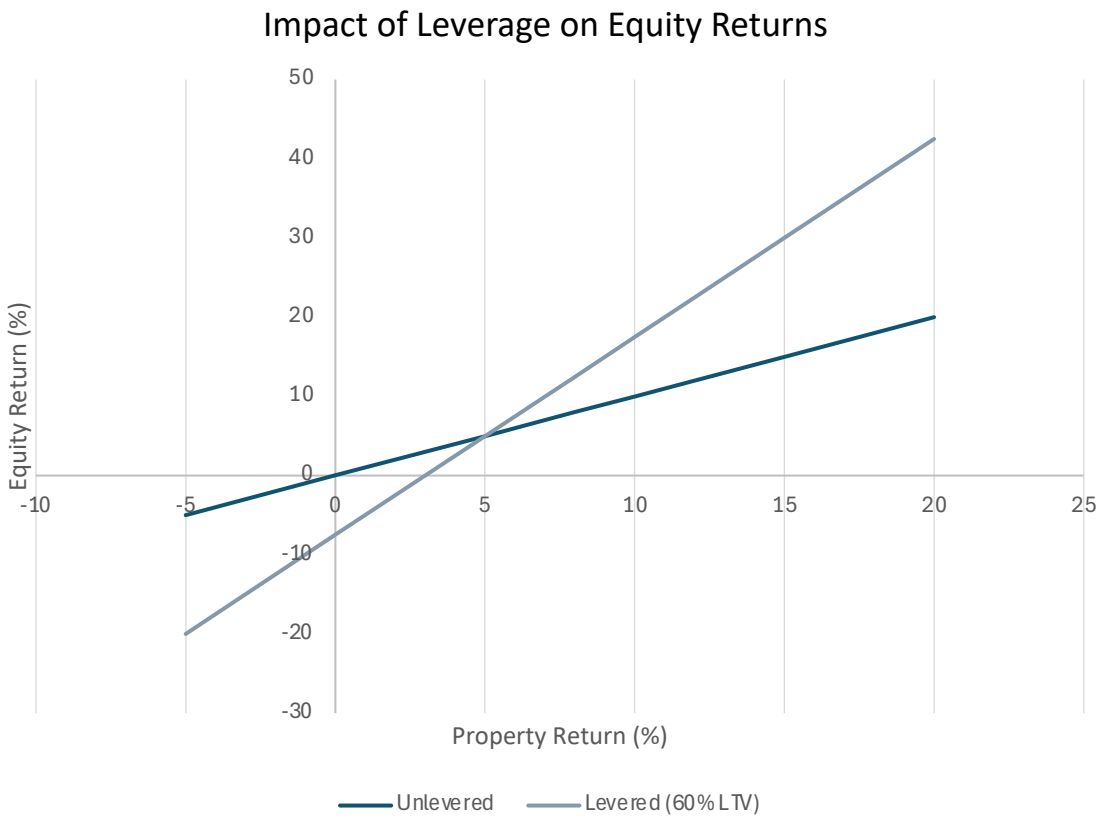
Q: What is the Relationship between Leverage and Return?

- Leverage increases equity return when property's returns exceed the cost of debt (5% in this example), as gains are earned on a smaller equity base
- At the breakeven point (property return = cost of debt), levered and unlevered returns are equal
- When property returns fall below the cost of debt, leverage amplifies losses due to fixed interest obligations

For example, to the right:

- Unlevered line moves one-for-one with property performance, whilst the line showing levered returns is amplified both below and above the 5% cost of debt
- Slope difference proves increased return volatility under leverage

Takeaways	• Leverage increases return volatility • Positive spread → return enhancement • Negative spread → equity erosion



ESCP Alternative Investment Society

Section IV: Case Studies

Real Estate Return Fundamentals

Without Calculator!

Core Formulas to use

Cap Rate:

$$CapRate = \frac{NOI}{Value}$$

Exit / Entry Value:

$$Value = \frac{NOI}{Cap\ Rate}$$

Equity Multiple:

$$EM = \frac{Total\ Cash\ In}{Equity\ Invested}$$

Cash-on-Cash:

$$CoC = \frac{Annual\ CashFlow}{Equity}$$

What Interviewers Evaluate

- Fluency in core valuation mechanics (NOI ↔ Cap Rate ↔ Value)
- Pricing intuition and yield sensitivity awareness
- Speed and accuracy in mental calculations
- Structured thinking under time pressure

Preparation Guidance

- Practice dividing by 4%, 5%, 6% quickly
- Memorize common valuation shortcuts (e.g., 5% = ×20, 4% = ×25)
- Train the IRR shortcut, for when there are no interim cashflows:
  EM^(1/n) − 1
- Get comfortable structuring 5-year cash flows on paper
- **Always verbalize your logic while calculating!**

T1: Basic Acquisition

Valuation Fundamentals - Cap Rate Mechanics

Case:
You are evaluating the acquisition of an office building generating an annual NOI of €10m.
The market entry cap rate for comparable assets is 5%.

Q1:	What is the implied purchase price?

Answer:

- Purchase Price: $\frac{10m}{5\%} = €200m$

Q2:	If market cap rates compress to 4%, what would be the new market value of the asset?

Answer:

- Market Value: $\frac{10m}{4\%} = €250m$

Key Insight:
Lower cap rate → Higher valuation. Small yield changes have large price impact.

Difficulty: ⭐

T2: Reverse Cap Rate Logic

Understanding Implied Yield

Case:
You are offered a stabilized property generating €10m NOI.
The seller is asking €220m.

Q1:	What cap rate are you implicitly paying?

Answer:

- Cap Rate: $\frac{10m}{220m} = 4.55\%$

Q2:	If market cap rates are 5%, are you paying a premium or a discount?

Answer:

- Cap Rate: 4.55% < 5.0% → Paying Premium

Key Insight:
Price ↑ → Yield ↓: Buying at 4.55% in a 5% market = Paying a premium.

Difficulty: ⭐⭐

T3: Capex Value Creation

Yield-on-Cost vs Market Yield

Case:
You acquired a building for €100m at a 5% cap rate. You are considering a €15m capex program expected to increase NOI by €1m annually. Market cap rates remain at 5%.

Q1:	What is the yield on cost of the capex?

Answer:

- YoC: $\frac{1m}{15m} = 6.67\%$

Q2:	Does this investment create value?

Answer:

- New NOI: €6m
- Value: $\frac{6m}{5\%} = €120m$, Value Increase = €20m
  → Net Value Creation = +€5m

Key Insight:
Invest when Yield-on-Cost > Market Cap Rate. This is yield arbitrage.

Difficulty: ⭐⭐

T4: Yield Compression (Unlevered)

Impact of Cap Rate Movements

Case:
You acquire a stabilized asset generating €10m NOI at a 5% cap rate. You hold the asset for 5 years. At exit, cap rates compress to 4.5%. NOI remains constant.

Q1:	What are the purchase price and exit value?

Answer:

- Entry: $\frac{10m}{5\%} = €200m$
- Exit: $\frac{10m}{4.5\%} = €222.2m$

Q2:	How does yield compression affect the equity multiple?

Answer:

- 5-year NOI: €50m
- Total Cash: €272.2m
- EM: $\frac{272.2m}{200m} = 1.36x$

Key Insight:
Compression = Multiple Expansion; Terminal value drives returns.

Difficulty: ⭐⭐

T5: Yield Expansion (Unlevered)

Cap Rate Risk

Case:
You acquire a stabilized asset generating €10m NOI at a 5% cap rate. You hold the asset for 5 years. At exit, cap rates expands to 5.5%. NOI remains constant.

Q1:	What is the exit value?

Answer:

- Entry: $\frac{10m}{5.0\%} = €200m$
- Exit: $\frac{10m}{5.5\%} = €181.8m$

Q2:	Why does IRR decline despite stable NOI?

Answer:

- 5-year NOI: €50m
- Total Cash: €231.8m
- EM: $\frac{231.8m}{200m} = 1.16x$
- Entry vs. Exit $200m - 181.8m = €18.2m$ Capital loss

Key Insight:
Cap rate expansion destroys equity value. Real estate behaves like a duration asset.

Difficulty: ⭐⭐⭐

T6: Introducing Leverage

Impact of Debt on Equity Returns

Case:
You acquire the €200m asset using 50% LTV debt at 4% interest (interest-only).
NOI remains €10m annually.

Q1:	What is the annual cash flow to equity?

Answer:

- Debt: $200m \times 50\% = €100m$
- Interest: $100m \times 4\% = €4m$
- Cash Flow to Equity: $10m - 4m = €6m$

Q2:	What is the cash-on-cash return?

Answer:

- CoC: $\frac{6m}{100m} = 6\%$

Key Insight:
Leverage enhances returns when asset yield exceeds cost of debt.

Difficulty: ⭐⭐⭐

T7: Compression With Debt

Leverage and Upside Convexity

Case:
You acquire a stabilized asset generating €10m NOI at a 5% cap rate (€200m value). The acquisition is financed with 50% LTV interest-only debt at 4%. You hold the asset for 5 years. At exit, cap rates compress to 4.5%.

Q1:	What is the equity value at exit?

Answer:

- Exit Value: $\frac{10m}{4.5\%} = €222.2m$
- Debt Repaid: €100m
- Equity at Exit: $222.2m - 100m = €122.2m$
- Operating Cash Flow: $6m \times 5yrs = €30m$
- Total Equity Cash: $30m + 122.2m = €152.2m$
- Equity Multiple: $\frac{152.2m}{100m} = 1.52x$

Extra: IRR $\approx 1.52^{(\frac{1}{5})} - 1$ $\approx$ 8.5% - 9% (not exact due to interim cashflows)

Q2:	Why does leverage increase IRR in this scenario?

Answer:
Investment Principle

- Debt is fixed at €100m
- Asset value increases due to cap rate compression
  - The entire valuation uplift accrues to equity
- Leverage therefore amplifies upside convexity when:
  Asset Yield > Cost of Debt
  AND
  Exit value increases.

Difficulty: ⭐⭐⭐⭐

T8: Expansion With Debt

Leverage and Downside Amplification

Case:
You acquire a stabilized asset generating €10m NOI at a 5% cap rate (€200m value). The acquisition is financed with 50% LTV interest-only debt at 4%. You hold the asset for 5 years. At exit, cap rates expand to 5.5%.

Q1:	What is the equity value at exit?

Answer:

- Exit Value: $\frac{10m}{5.5\%} = €181.8m$
- Debt Repaid: €100m
- Equity at Exit: $181.8m - 100m = €81.8m$
- Operating Cash Flow: $6m \times 5yrs = €30m$
- Total Equity Cash: $30m + 81.8m = €111.8m$
- Equity Multiple: $\frac{111.8m}{100m} = 1.12x$

Extra: IRR $\approx 1.12^{(\frac{1}{5})} - 1$ $\approx$ 2.3% - 2.5% (not exact due to interim cashflows)

Q2:	Why is the downside amplified?

Answer:
Investment Principle

- Debt stays fixed at €100m
- Asset value declines due to cap rate expansion
  - The decline in value is fully absorbed by equity
- Leverage increases return volatility because:
  - Debt is fixed
  - Asset value fluctuates
  - Equity takes the shock

Difficulty: ⭐⭐⭐⭐

T9: Financing Structure Choice

IRR Optimization vs Risk Management

Case:
You are evaluating two financing structures for a 5-year hold: an Amortizing Loan or an Interest-Only Loan.
Both have identical interest rates.

Q1:	Which structure maximizes IRR?

Answer:
- Interest-Only Loan:
  - Higher annual cash flow
  - No principal amortization
  - More early cash to equity

→ IRR increases due to earlier cash distributions

Q2:	Which structure reduces risk?

Answer:
- Amortizing Loan:
  - Lower exit debt balance
  - Reduced refinancing risk
  - Lower leverage over time

→ IRR rewards early cash flow
→ Risk management Favors gradual deleveraging

Difficulty: ⭐⭐⭐⭐

T10: Development Timing & IRR

Timing Effects in Value Creation

Case:
You are planning a €20m development program expected to increase NOI by €3m annually once completed. You can execute the development: I) Immediately after acquisition OR II) Shortly before exit.

Q1:	When should you execute development to maximize IRR mathematically?

Answer:
- Analytical Logic
  - IRR Favors later value spikes
- Executing development just before exit generally:
  - Maximizes terminal value
  - Minimizes time capital is tied up

Q2:	Why might this not be optimal in reality?

Answer:
Although late development maximizes IRR mathematically, real-world constraints matter:
- Leasing risk
- Construction risk
- Market cycle timing
- Capital markets conditions

→ IRR is highly timing-sensitive,
but real estate investing is risk-adjusted

Difficulty: ⭐⭐⭐⭐

ESCP Alternative
Investment Society